LESSON PLAN · GENERATIVE AI (GR 6–8) INTERMEDIATE: GEN-AI CREATORS · LEVEL 2

Lesson 1: AI as a Coding Buddy

Module 1 | 60 minutes | Explain, don't write | Grades 7–11 · ages 12–16

Objective: Ask an AI to explain code line by line instead of writing it for you, then run the code yourself to catch the AI's mistakes.

Success criteria

- I can ask an AI to explain code line by line without rewriting it.
- I can run code to check an AI's explanation.
- I can explain a line of code in my own words.
- I can tell good buddy use from ghost-writing.

Key vocabulary

- **program**: A list of instructions a computer follows, in order.
- **line by line**: Going through code one line at a time.
- **loop**: Code that repeats, such as for r in runs.
- **variable**: A named box that stores a value, like total.
- **output**: What the program shows when it runs.
- **ghost-writing**: Getting someone (or AI) to do your work and handing it in as yours.

You will need

- Projector or screen share with Kitabash and the AI tools for this lesson
- Devices: 1 per learner (laptop, tablet or phone)
- Learner accounts only where allowed: see the course safety notes
- Printed worksheet (optional)
- Project files: l01-explain-my-code-kit.md, l01-teacher-demo-script.md

Timed lesson plan (60 min)

Time	Stage	What happens
5 min	**Warm-up game**	Human compiler game: one learner runs the cricket code out loud while the class predicts each total.
5 min	**Hook + thumbs check**	Show print("2" + "2"). Thumbs up if you think it prints 4, down for 22. Run it and reveal 22 — computers follow rules, not guesses.
5 min	**Watch**	Play the lesson video with captions; pause on the caught mistake and ask "How did we know the AI was wrong?"
8 min	**Live demo**	Teacher demo: run the code, ask the AI helper for a line-by-line explanation, catch any slip, then ask the AI to quiz the class.
8 min	**Prompt Lab (pairs)**	Pairs write the explain-my-code prompt, press Check and improve it until every item is ticked.

10 min	**Follow-along + log**	Everyone builds their explain-my-code log in the Python Lab and AI helper (13+ may use ChatGPT, Gemini or Copilot with permission).
6 min	**Quick competition**	Buddy or ghost-writer sort race, then Check the AI: range(3). Fastest all-correct trio gets a shout-out.
5 min	**Show-and-tell**	Two learners share one line they explained in their own words and one AI mistake they caught.
4 min	**Breakout: the buddy routine**	In small breakout rooms, order the buddy routine together and agree on the rule “run it first”.
4 min	**Probe + exit ticket**	New situation: the invented command, Explain one line, then the exit ticket in the chat.

Unplugged warm-up

Human compiler game (5 min): put RUNS on the board. One learner is “the computer” and must follow it exactly, saying each value of total out loud. The rest of the class are “AI explainers” who shout what they think happens next. Who was right — the explainers or the computer? Point out that the computer (running the code) is always the final judge.

Teaching the idea

Level 1 taught prompting and checking. Level 2 turns AI into a coding coach. The big rule: AI explains, hints and quizzes — the learner writes the code. A program is only “yours” if you can explain every line. AI explanations of code can contain slips (wrong totals, wrong loop counts, invented functions), so the learner always runs the code and compares. Repeat the privacy rule for code: never paste code containing passwords, keys or real people’s details. Age rules stay the same: under 13 use the Kitabash AI helper and the teacher’s screen; 13–17 may use ChatGPT, Gemini, Copilot or Perplexity with a parent’s permission; Claude is 18+ only.

Analogy: An AI coding buddy is like a cricket coach in the nets: they can explain your stance and point out mistakes, but they cannot bat for you in the match.

Live demo: Open the Kitabash Python Lab and the Kitabash AI helper side by side on the shared screen. 1) Type the cricket over tracker and press Run — circle “Total runs: 17”. 2) Paste it into the helper with: “Explain this Python code line by line for a Grade 7 student. Use simple words. Do not change or rewrite the code.” Read the reply aloud, pausing at each line. 3) Ask the class: “Does the total it gives match our output?” If the AI says 16 (or anything not 17), celebrate the catch. If it says 17, change the list to [4, 6, 1, 0, 6, 2] and ask the AI what it prints now — check it again. 4) Show the follow-up “Ask me two questions to check I understand.” Answer one wrongly on purpose and see how it responds.

Questions to ask

Ask	Listen for

What is the difference between a coding buddy and a ghost-writer?	A buddy explains and hints so you learn; a ghost-writer does the work for you.
How do you check an AI's explanation of code?	Run the code and compare the real output with what the AI said.
What should you never paste into an AI with your code?	Passwords, keys, real names, phone numbers or other personal details.
Why ask the AI to quiz you?	Answering in your own words shows you really understand the code.
What does NameError often mean when you used an AI's suggestion?	The name or command does not exist — the AI may have invented it.

Differentiation

Support
Give the explain prompt on a card with the blanks filled in. Pair learners so one runs the code while the other reads the AI explanation aloud.

Stretch
Ask the AI to explain the code "for a 6-year-old" and then "for a Class 12 student". Compare: which explanation is more accurate, and which is easier? Find one thing each version gets wrong or leaves out.

Common misconceptions

Misconception	How to fix it
If the AI explained it, I understand it	Reading is not understanding. Explain it back in your own words and answer the AI's quiz.
AI never makes mistakes about code	It can get loop counts, totals and command names wrong. Running the code is the test.
Using AI for code is always cheating	Asking for explanations and hints is fine; handing in code you did not write and cannot explain is not.
The longer the AI answer, the better	Short, line-by-line answers you can check are better than long essays.

Assessment

Use the Prompt Lab results (auto-checked), the quick checks and the Probe answers. Written answers arrive in your review queue.

Exit ticket
How do you know if an AI's explanation of your code is correct?

Homework
Pick any short program from your school textbook or Level 1 notes. Add three lines to your explain-my-code log and one AI mistake (or "no mistakes found — here is how I checked").

Answer key — Kitabash activities

1. Prompt Lab: explain my code (perform · ai lab)

Checklist: Ask for line by line · Say your level · Ask for simple words or an example · Tell it not to rewrite the code · Ask it to check your understanding

Strong example:

```
You are a patient coding coach. I am a Grade 8 beginner learning Python. Explain my code line by line in simple words, with an everyday example for the loop. Do not change or rewrite the code. At the end, ask me two questions to check I understand. Here is my code: [my code]
```

2. Coding buddy or ghost-writer? (perform · categorize)

Good buddy use: “Explain what line 4 of my code does.”, “Quiz me on how this loop works.”, “What does this error message mean?”, “Give me a hint, not the answer.”
Ghost-writing (not OK): “Write my whole homework program for me.”, Copying AI code and saying you wrote it, Pasting the teacher’s test question and submitting the answer

3. Check the AI: range(3) (perform · code output)

4. The buddy routine (perform · ordering)

Order: Run the code yourself and note the output → Ask the AI to explain it line by line → Compare what the AI said with the real output → Write each line in your own words → Ask the AI to quiz you

5. New situation: the invented command (probe · multiple choice)

Answer: The AI invented a command that does not exist in Python

6. Explain one line (probe · free response)

Look for: add, total, run, check. **Example:** It takes the total so far and adds the next number r to it, then stores the new total. To check the AI, I would run the code and see if the printed total matches the sum I work out myself.

Answer key — worksheet

1. **Key words:** program = A list of instructions a computer follows, in order. · line by line = Going through code one line at a time. · loop = Code that repeats, such as for r in runs. · variable = A named box that stores a value, like total.
2. **Prompt Lab: explain my code:** Include: Ask for line by line; Say your level; Ask for simple words or an example; Tell it not to rewrite the code; Ask it to check your understanding. Example: You

are a patient coding coach. I am a Grade 8 beginner learning Python. Explain my code line by line in simple words, with an everyday example for the loop. Do not change or rewrite the code. At the end, ask me two questions to check I understand. Here is my code: [my code]

3. **Coding buddy or ghost-writer?:** Good buddy use: “Explain what line 4 of my code does.”, “Quiz me on how this loop works.”, “What does this error message mean?”, “Give me a hint, not the answer.” · Ghost-writing (not OK): “Write my whole homework program for me.”, Copying AI code and saying you wrote it, Pasting the teacher’s test question and submitting the answer
4. **Check the AI: range(3):** A. Hi three times (no numbers)
5. **The buddy routine:** 1 Run the code yourself and note the output · 2 Ask the AI to explain it line by line · 3 Compare what the AI said with the real output · 4 Write each line in your own words · 5 Ask the AI to quiz you

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